Lukas Liakhovitch

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EDUCATION

Purdue University, *B.S. in Mechanical Engineering* (Aug 2025 - May 2028), West Lafayette, IN
GPA 3.93/4, Dean's List & Semester Honors Fall 2025, Alpha Lambda Delta Phi Eta Sigma Honor Societies

Pope John XXIII Regional High School, Sparta, NJ (Aug 2021 - May 2025), *AP Scholar with Distinction*

EXPERIENCE

Active Controls Structures and Manufacturing Team Member (January 2026 - Present)
***PSP**: Purdue Space Program, West Lafayette, IN*

- Designed and manufactured structural members and fluid system mounts for a 9 ft, 500 lb thrust-vectoring self-landing rocket in NX
- Performed FEA in NX SimLab and ANSYS on lander components, iterating designs to meet structural requirements

Software Engineering Intern (May 2025 - Present)
***Clearcut Software**, Fair Lawn, NJ*

- Ported financial processing library (Pyxirr) from Rust to C++ and created custom DuckDB extension for its use
- Developed SQL and Python unit tests to validate extension functionality and format results for presentation

Torch Igniter Rocket Engine & Turbopump Development Team Member (September 2025 - January 2026)
***PURPL**: Purdue Undergraduate Rocket Propulsion Lab, West Lafayette, IN*

- Design, CAD, and manufacturing of central shaft bearing housing for first undergrad student-developed turbopump
- Design and machining of a custom torch igniter rocket engine (team of 4)

Founder & Captain (April 2023 – June 2025)
***FIRST Tech Challenge (FTC) Robotics - Ultraviolet 23268**, Hackettstown, NJ*

- Founded, organized, recruited, fundraised for, and managed award-winning NJ State Champion team, ranked #4 (out of 8,000+ globally) in its 2nd year, after competing at FIRST World Championship in 2024-25 season.
- Achieved 2nd highest average score globally and one of the most consistent autonomous routines in the world
- Self-directed on a 3-person independent team (typical teams have 10-15) with no institutional backing and a limited budget, while mentoring two junior members from no prior experience
- Created in Fusion360 and assembled 18x18 in. robot; implemented custom pathfinding and control algorithms (PID, motion profiling, inverse kinematics) in Java to optimize autonomous navigation, focusing on consistency

Scholar (Sept 2024 – May 2025)
***NJ Governor's STEM Scholars Program,** NJ (10% acceptance rate)*

- Learned about STEM in academia, government, and professional world; completed and presented a research project on waste management with a small team of peers; engaged and networked with STEM professionals

Design Intern (June 2024 – Aug 2024)
***L&M Architectural Signs**, Boonton, NJ*

- Assisted with architectural design of a 25-foot-tall rotating sign project using Fusion 360
- Worked with fabricators and engineers to CAD several prototype drive systems for a project

PROJECTS

Restoration and Repair (May 2024 – Aug 2025)

- Repaired/restored a used vehicle by diagnosing issues and replacing worn or missing components (exhaust system, radiator, brakes, and audio system); repaired laser cutter machine by replacing command module and fixing localization/control issues.

SKILLS & VOLUNTEERING

TECHNICAL: Java, C, MATLAB, ANSYS, Python, Git, Linux/WSL, Fusion360, Onshape, NX/SimLab, machining, fabrication
SOFT: Project Management, Team Leadership, Communications, Advocacy, Problem-solving, Root-cause Analysis